

ARYAN VASHISHTH

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Summary

Full-Stack Engineer building real-time backends for AI-powered products. At Stimuler, shipped Conversation V3 end-to-end (**94% p50 latency reduction**), real-time voice calling on **Gemini Live**, and **multi-LLM orchestration** integrating 4+ providers. Strong in TypeScript/Node.js, distributed systems, real-time/voice infrastructure, and cross-team delivery.

Education

Thapar Institute of Engineering and Technology

May 2022 – May 2026

BE, Computer Engineering, CGPA: 8.16/10

Patiala, Punjab

Skills

Languages: C++, Java, JavaScript, TypeScript, SQL, Dart

Computer Science: Data Structures & Algorithms, Object-Oriented Design, System Design, Operating Systems, DBMS, Distributed Systems, Computer Networks

Backend: Node.js, Bun + Hono, Express.js, REST APIs, WebSockets, Microservices, Caching, Redis

Frontend: React.js, Next.js, Flutter, HTML, CSS, Responsive Design

Databases: PostgreSQL, MongoDB, Firestore, Redis

Cloud & DevOps: AWS (EC2, ECS, Lambda, S3), Firebase, Docker, Nginx, CI/CD, GitHub Actions, Git

Certifications: AWS Certified AI Practitioner (AIF-C01); Blockchain by IIT Kanpur

Experience

BASETHESIS LABS | Software Engineer, Contract

Jun 2026 – Present

- Owned **Synth's TallyPrime integration** for an **AI CFO (76 users)**, kayess.chat.getsynth.io. Unlike Zoho/Microsoft/ERPNext, **Tally has no API**; reverse-engineered its XML protocol.
- Built a **deterministic financial-statement engine** (P&L, BS, TB, aging, TDS, GST); **matched Tally** on 15 client books.
- Worked directly with client **CFOs and financial analysts**; reverse-engineered how they run their books, shipped and validated the automation on their live accounts.
- Turned Tally's flaky single-threaded XML gateway into a reliable production write path with a **half-open circuit breaker**, single-flight mutex, and **month-windowed batched reads** (byte + wall-clock caps); root-caused and eliminated a reproducible **OOM wedge** from unbounded full-history exports.
- Built the **AI write-path safety model: REMOTEID** identity, fail-closed dedup, phantom-write detection, and resumable UNKNOWN-state bulk writes; agents post into live client books without double-posts.

STIMULER | Full-Stack Engineer Intern

Jan 2026 – Jun 2026

- Cut Conversation V3 p50 latency **23s** → **~1.4s (~94%)** by re-architecting backend off Firebase Cloud Functions onto **Bun + Hono** for native WebSocket streaming; drove the system to upstream-bound via 2 cross-team optimizations.
- Rebuilt voice calling server-side on **Gemini Live + Bun + Hono** (client-side Hume couldn't capture audio); shipped **24kHz PCM streaming**, 500ms barge-in stitching, **dynamic RMS noise gating**; killed Gemini's post-2-min latency cliff; rolled out to **~3K users** via SHA-1 bucketing.
- Architected V3's **multi-LLM runtime** with **4+ providers** (OpenAI, GPT-4o, in-house on **Modal + vLLM**), swappable live via Firestore config; engineered **Qwen3 output gates** blocking regressions without latency cost.
- Built calling-ops dashboard solo (**Flutter Web**, Firebase Auth); shipped a **cross-session memory layer** that LLM-summarizes voice + chat into per-user profiles threaded into every prompt.
- Migrated **ERS pronunciation drill scoring** to Stimuler's in-house model (off Microsoft's scoring API), preserving full real-time scoring.
- Shipped **self-healing V2→V3 migration** (lazy, per-user) across **150+ topics**; zero downtime, **40+ Jest tests**.

ELC | Research and Engineering Intern, On-Site

Jun 2025 – Jul 2025

- Led a **team of 5** to build a scalable **TGNN-based Ethereum fraud detection pipeline** processing **3M+ transactions** end-to-end (data ingestion through prediction API).
- Engineered **8 wallet-level features** and migrated to **GCN + LightGBM**, improving accuracy and recall by **30%**.

Projects

Real-Time Distributed Exchange Engine | TypeScript, Node.js, Redis | [GitHub](#)

- Built a heap-based order book (**$O(\log n)$** insertion + best-price matching) and an event-driven matching engine on **Redis Pub/Sub**; sustained **10,000+** orders with partial fills and real-time trade ID generation.
- Built an async order-processing pipeline with execution lifecycle tracking (open → matched → filled); **non-blocking event loop** maintained consistent match latency under concurrent order submissions.

ShopXar | React.js, TypeScript, Node.js, MongoDB, Three.js, AR/3D | [Live Demo](#)

- Built a **3D e-commerce platform with AR previews** (Three.js, model-viewer); cut model-load latency **40%** via CDN caching, achieved **100% cross-device compatibility**.

Achievements

Co-authored research paper on **TGNN-based Ethereum fraud detection** (under review) · **Runner-up**, AIESEC Hackathon · **Best Speaker & Narrator**, TEDxTIET + Echoes Club.